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Preface

Why an Analyst's Companion? Millions of analyses are carried out every day in laboratories for all sectors of industry and science. Many people are willing to pay for these analyses because they are considered effective in making a scientifically sound decision. Though few publications address the economics of analytical sciences, nonetheless, a report by the European Commission concluded in 2002 that "for every euro devoted to measurement activity, nearly three euros are generated" [1]. But is it easy and simple to use an analytical result, and does it always allow you to make the right decision? Some questions illustrate the risks involved in relying on a result:

- How do you know that the laboratory used the method that gave the exact result?
- Like any measurement, analysis is subject to error. How can you estimate them?
- How can a spurious measurement be used effectively?

This is the right time to explain why and how the concept of measurement uncertainty (MU) can be used to better manage these risks. This also means that a new challenge for analysts is to develop an appropriate method for estimating MU more explicitly applicable to analytical sciences. In this perspective, a tool based on the statistical dispersion intervals called method accuracy profile (MAP) is proposed as the backbone of the book. The theoretical aspects of the MAP procedure and MU estimation are presented in several examples and template worksheets to help analysts quickly grasp this tool.

At the turn of the 1970s, three analytical chemists, Bruce Kowalski, Luc Massart and Svante Wold, conceptualized a discipline they called Chemometrics [2]. Unfortunately, they all have passed away since, but their work is still vivid. Many chemometrics books have been published, proving the added value of statistics to analytical sciences. Some are globally addressing chemometrics [3–5] other are more focused on statistics [6, 7], and others on method validation [8, 9].

This book contributes to the application of chemometrics, but the obvious aim is not to repeat what is available in many valuable publications. Only a few books precisely address measurement uncertainty in analytical sciences [10–12]. They present limited facets and do not propose a more comprehensive approach. The aim of this book is to describe a global procedure for MU estimation, easily applicable in analytical laboratories. In a recent publication, we have exposed in a condensed manner

our view of the link between validation and measurement uncertainty [13]. This book develops more extensively and practically our viewpoint.

However, it is not satisfactory to simply propose a *modus operandi* (even if it is claimed to be universal) for estimating MU when this parameter is still new in analytical sciences and not always well identified by end-users. Therefore, several chapters are dedicated to its practical use in decision-making, demonstrating its advantages. These remarks indicate that this book is primarily intended for professional analysts, although researchers and students may find it of interest.

In order to reach this goal, the book is organized around practical responses covering three major questions daily put to analysts when they develop a new method or routinely apply it to unknown samples:

- How to quantify the analyte?
- How to validate the method?
- How to estimate the measurement uncertainty?

How does this book give answers these questions? We use as a roadmap a tool based on the application of statistical dispersion intervals called MAP. The latter was initially conceived for method validation, but it can easily be used for MU estimation. While method validation is often reduced to computing a set of disconnected parameters to be estimated, the MAP approach is more global. It consists in defining the interval where the method is able to produce a given proportion of acceptable results. This perspective is in harmony with the uncertainty approach proposed by metrologists some decades ago that consists in computing the so-called coverage interval of the result.

The chapters of the book can be read independently. This may explain some redundancies in the quoted publications. But they are structured according to a reading thread illustrated in Figure 1. The thick grey arrow is the backbone. Six main chapters are characterized as rounded angle boxes. Three of them are devoted to measurement uncertainty, as it is a key issue of the book.

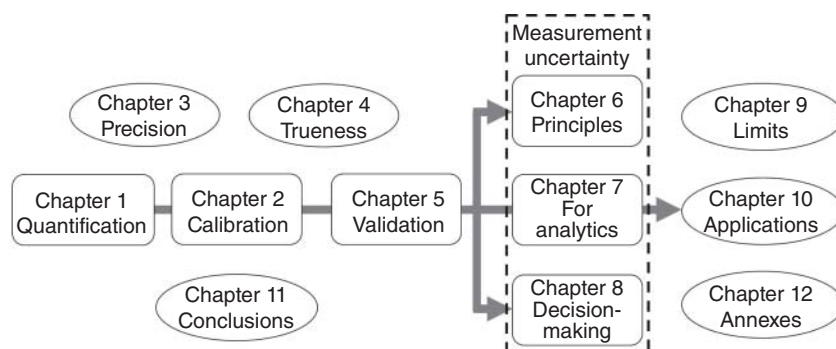


Figure 1 How to read this book.